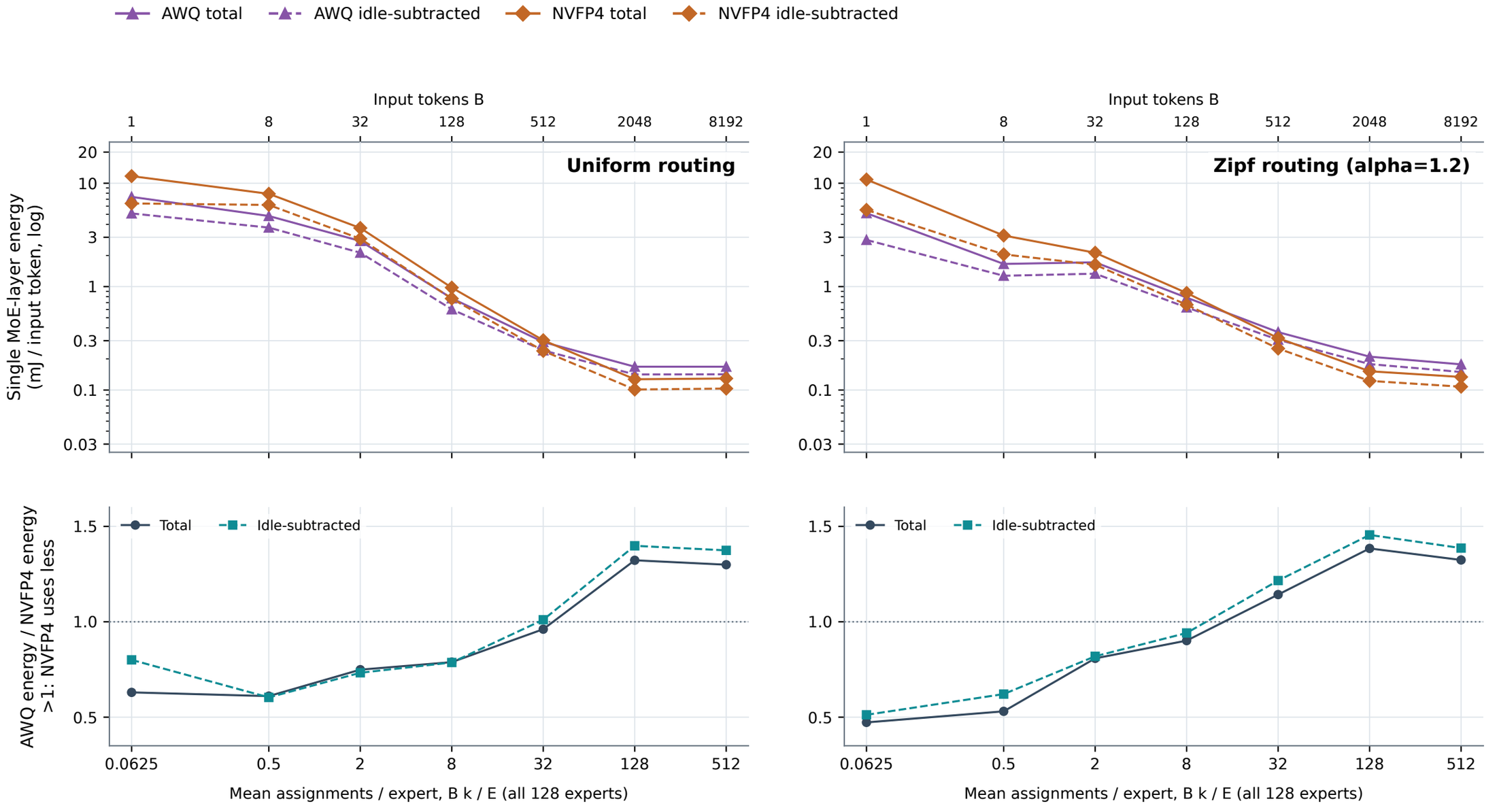
Qwen3 calibrated low-bit device energy

COMPLETE: 28/28 energy windows

AWQ W4A16 Marlin vs NVFP4 W4A4 CUTLASS | Energy protocol v2 only | Same expert shape/routes



Only matched energy v2 is compared. Older synthetic BF16/FP8 energy v1 is excluded.

NVML counter after idle baseline and continuous-load warmup; one ~3 s window per point. Small energy gaps remain unconfirmed.

Two public checkpoints; not same weights or whole-model J/token. No repeated-window error bars, quality or serving claim.

Ratios use measured total and idle-subtracted energy separately; no crossover interpolation. Both low-bit formats observed at all 14 routing/load points.